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Classification and Logistic Regression

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Binomial Logistic Regression (1)

- Structure of a generalized linear model

  – Comparison with linear regression model

$$LRM: \ E[Y|x] = \mu_{Y|x} = X\beta = \sum_{i=0}^{m} \beta_i X_i$$

$$GLM: \ E[Y|x] = \mu_{Y|x} = \psi^{-1}(X\beta)$$

$$\psi(\mu) = X\beta$$

ψ : 연결함수(Link Function)

  – If the link function is Logit, the dependent variable is expressed in binary (0 and 1).

  Logistic regression model applicable in capable cases

$$log\left(\frac{p_i}{1-p_i}\right)$$

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Binomial Logistic Regression (2)

- Assuming the dependent variable follows a binomial distribution:

  – Dependent variable to be estimated: (Y=1) probability

$$\psi(\mu) = \log \frac{P(Y=1|X_1, X_2, \cdots, X_p)}{1 - P(Y=1|X_1, X_2, \cdots, X_p)} = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p$$

$$\hat{p}(Y=1|X_1, X_2, \cdots, X_p) = \frac{\exp(\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p)}{1 + \exp(\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p)}$$

(The right-hand side is the sigmoid function)

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Binomial Logistic Regression (3)

- If the composition of the exponential function (exp) is set as a linear expression,

$$log\left(\frac{p_i}{1-p_i}\right) = \beta_0 + \beta_1 x_1$$

– Regression coefficients using simple linear regression analysis ÿ¡It is possible to estimate,

Multiple linear regression analysis is possible even when constructed as a polynomial.

– Critical value for determining 1 and 0 for the estimate of y in the constructed regression model

The (threshold/cut-off value) is commonly set to 0.5, but

All values between (0, 1) are possible.

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Multinomial Logistic Regression Analysis (1)

- Multinomial Logistic Regression

  – The value of the dependent function (y) is not (0, 1) but like (0, 1, , 2, 3, ...), 2 of which are

  It can be utilized when divided into upper classes.

  – Extending the binomial logistic regression model to multinomial logistic regression

  – Multinomial Logistic Regression

  - In Logit Link, it is not the ratio of success-failure odds, but the probability of being a specific class

    Use the ratio of the probability of being of a certain class relative to the rate.

  - If the problem has K classes, estimate (K-1) linear regression equations

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Multinomial Logistic Regression (2)

- Multinomial Logistic Regression
  - In Logit Link, not the ratio of success to failure odds, but the probability of being a specific class versus Use the ratio of probabilities of a certain class
  - If the problem has K classes, estimate (K-1) linear regression equations

$$\ln\frac{P(Y=1)}{P(Y=K)}=X\beta^{(1)},\ \ln\frac{P(Y=2)}{P(Y=K)}=X\beta^{(2)},\ \cdots,\ \ln\frac{P(Y=K-1)}{P(Y=K)}=X\beta^{(K-1)}$$

  - Method to obtain posterior probabilities from the above (K-1) regression equations (composed of sigmoid functions)

$$P(y_i=K)=1\Big/\left(1+\sum_{i=1}^{K-1}\exp\left(\beta^{(i)}X_i\right)\right)$$

$$P(y_i=1)=\exp\left(\beta^{(1)}X_i\right)\Big/\left(1+\sum_{i=1}^{K-1}\exp\left(\beta^{(i)}X_i\right)\right)$$

$$\cdots$$

$$P(y_i=K-1)=\exp\left(\beta^{(K-1)}X_i\right)\Big/\left(1+\sum_{i=1}^{K-1}\exp\left(\beta^{(i)}X_i\right)\right)$$

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Multinomial logistic regression analysis (3)

- Utilization of the softmax function

  – In the equations below, instead of the constant term '1' on the right-hand side ' $\exp\left(\beta^{(K)} X_i\right)$ If you enter ',

$$P(y_i = K) = 1 / \left(1 + \sum_{i=1}^{K-1} \exp\left(\beta^{(i)} X_i\right)\right)$$

$$P(y_i = 1) = \exp\left(\beta^{(1)} X_i\right) / \left(1 + \sum_{i=1}^{K-1} \exp\left(\beta^{(i)} X_i\right)\right)$$

$$\cdots$$

$$P(y_i = K-1) = \exp\left(\beta^{(K-1)} X_i\right) / \left(1 + \sum_{i=1}^{K-1} \exp\left(\beta^{(i)} X_i\right)\right)$$

$$P(y_i = \boldsymbol{j}) = \exp\left(\beta^{(j)} X_i\right) / \sum_{i=1}^{K} \exp\left(\beta^{(i)} X_i\right)$$ (j = *1, … ,K)* * softmax function

  – Using the softmax function instead of the sigmoid function for multiclass problems, multinomial logistic

  It is often used in regression analysis.

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Binomial Logistic Regression Analysis Practice

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Composition of the analysis model (1)

- Data preparation
  - Wisconsin breast cancer dataset
    - Binary classification data determining breast cancer (malignant tumor)
    - Many feature data required for judgment (binary classification)
  - Size, shape, etc. of the tumor
  - Configuration data (data: 569ÿ31, target: 569)

```
import pandas as pd
from sklearn.datasets import load_breast_cancer

b_cancer = load_breast_cancer()
b_cancer_df = pd.DataFrame(b_cancer.data, columns=b_cancer.feature_names)
b_cancer_df['diagnosis'] = b_cancer.target
print('* row number: ', len(b_cancer_df))
print('* column number: ', len(b_cancer_df.columns))
```

```
* row number:  569
* column number:  31
```

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Composition of the analysis model (2)

- Identifying feature information

```
b_cancer_df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 569 entries, 0 to 568
Data columns (total 31 columns):
 #   Column                   Non-Null Count  Dtype
---  ------                   --------------  -----
 0   mean radius              569 non-null    float64
 1   mean texture             569 non-null    float64
     ........
 29  worst fractal dimension  569 non-null    float64
 30  diagnosis                569 non-null    int32
dtypes: float64(30), int32(1)
memory usage: 135.7 KB
```

- Identifying and standardizing feature values

```
print(b_cancer.data[0])
from sklearn.preprocessing import StandardScaler  # 데이터 표준화 n(0, 1)
scaler = StandardScaler()
b_cancer_scaled = scaler.fit_transform(b_cancer.data)
print(b_cancer_scaled[0])
```

```
[1.799e+01 1.038e+01 1.228e+02 1.001e+03 1.184e-01 2.776e-01 3.001e-01
 1.471e-01 2.419e-01 7.871e-02 1.095e+00 9.053e-01 8.589e+00 1.534e+02
 6.399e-03 4.904e-02 5.373e-02 1.587e-02 3.003e-02 6.193e-03 2.538e+01
 1.733e+01 1.846e+02 2.019e+03 1.622e-01 6.656e-01 7.119e-01 2.654e-01
 4.601e-01 1.189e-01]
[1.09706398 -2.07333501  1.26993369  0.9843749   1.56846633  3.28351467
 2.65287398  2.53247522  2.21751501  2.25574689  2.48973393 -0.56526506
 2.83303087  2.48757756 -0.21400165  1.31686157  0.72402616  0.66081994
 1.14875667  0.90708308  1.88668963 -1.35929347  2.30360062  2.00123749
 1.30768627  2.61666502  2.10952635  2.29607613  2.75062224  1.93701461]
```

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Composition of the analysis model (2)

- Perform binomial logistic regression analysis using the prepared data
  - Import necessary modules
  - Set 'diagnosis' as Y and b_cancer_scaled scaled by a normal distribution as X.
  - Split the total 569 datasets into training and evaluation data in a 7:3 ratio.
  - Create Logistic Regression Model Object (lr_b_cancer)
  - Train the model using fit() with the training data
  - Predict the trained model using predict() with evaluation data (X_test)

```
from sklearn.linear_model import LogisticRegression
from sklearn.model_selection import train_test_split

Y = b_cancer_df['diagnosis']
X = b_cancer_scaled
X_train, X_test, Y_train, Y_test = train_test_split(X, Y,
                        test_size = 0.3, random_state = 100)
lr_b_cancer = LogisticRegression() # logistic 회귀모델 생성
lr_b_cancer.fit(X_train, Y_train)  # logistic 회귀모델 룬련
Y_predict = lr_b_cancer.predict(X_test) # 테스트 데이터러 예측 수행
```

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Performance evaluation of the configured model

- Verify the performance of the created model – Import necessary modules from the sklearn.metrics package – Calculate misclassification tables for test data split 7:3
  - TP:63, FP:0, FN:6, TN: 102

– Check accuracy (0.965), precision (0.944), recall (1.0), F1 (0.971), etc.

```
from sklearn.metrics import confusion_matrix, accuracy_score
from sklearn.metrics import precision_score, recall_score, f1_score,
                            roc_auc_score

print('* 오차행렬: ', confusion_matrix(Y_test, Y_predict))

accuracy = accuracy_score(Y_test, Y_predict)
precision = precision_score(Y_test, Y_predict)
recall = recall_score(Y_test, Y_predict)
f1 = f1_score(Y_test, Y_predict)
roc_auc = roc_auc_score(Y_test, Y_predict)
print('* 정확도: {0:.3f}, 정밀도: {1:.3f}, 재현율: {2:.3f}, F1: {3:.3f}'.
      format(accuracy, precision, recall, f1))

print('* ROC_AUC: {0:.3f}'.format(roc_auc))
```

```
* 오차행렬:  [[ 63   6]
             [  0 102]]
* 정확도: 0.965, 정밀도: 0.944, 재현율: 1.000, F1: 0.971
* ROC_AUC: 0.957
```

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Multinomial Logistic Regression Analysis Practice

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Data preparation

- iris data
  - 150 samples
  - Input: sepal_length, sepal_width, petal_length, petal_width – target: class (3 classes)

```
from sklearn import datasets
import pandas as pd
iris = datasets.load_iris()

df = pd.DataFrame(iris.data, columns=iris.feature_names)
df['class'] = iris.target

df.columns=['sepal_length', 'sepal_width', 'petal_length', 'petal_width', 'class'] df.head(5)
```

	sepal_length	sepal_width	petal_length	petal_width	class
0	5.1	3.5	1.4	0.2	0
1	4.9	3.0	1.4	0.2	0
2	4.7	3.2	1.3	0.2	0
3	4.6	3.1	1.5	0.2	0
4	5.0	3.6	1.4	0.2	0

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Training/validation data preparation and one-hot encoding

```
from sklearn.model_selection import train_test_split from
tensorflow.keras.utils import to_categorical

x_data=df.loc[:, 'sepal_length':'petal_width'] # Set input feature as range y_data=df.loc[:,
'class'] # Target feature

# Split 20% of the data sample for validation, and the remaining 80% for
validation x_train, x_valid, y_train, y_valid = train_test_split(x_data, y_data, train_size=0.8, random_state=1)

#one-hot encoding y_train =
to_categorical(y_train) y_valid =
to_categorical(y_valid)

# Print 5 at a
time print(y_train[:5]) ; print(y_valid[:5])
```

```
[[0. 1. 0.]
 [0. 0. 1.]
 [0. 1. 0.]
 [0. 0. 1.]
 [0. 0. 1.]]
[[1. 0. 0.]
 [0. 1. 0.]
 [0. 1. 0.]
 [1. 0. 0.]
 [0. 0. 1.]]
```

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Configuration and Execution of Analysis Model (1)

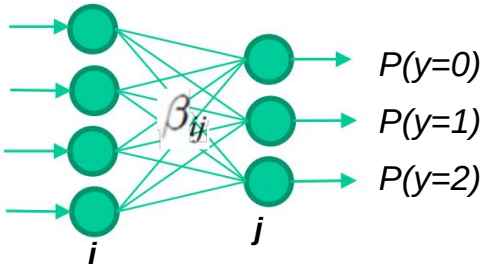
j

- Utilization of artificial neural networks using the Softmax function

  – An artificial neural network with 4 nodes (input features) in the input layer and 3 nodes in the output layer

  – Set the activation function to the softmax function

  – Optimizer (learning rate) setting methods: Adam, SGD, Adagrad, etc.



$$P(y=j)=\exp\left(\beta^{(j)}X_i\right)/\sum_{i=1}^{K}\exp\left(\beta^{(i)}X_i\right)$$

```
# Preparing TensorFlow

from tensorflow.keras import Sequential from
tensorflow.keras.layers import Dense, Activation

# Preparation and configuration of the
neural network model model
= Sequential() model.add(Dense(3, input_dim=4, activation='softmax')) # The number of input variables is 4 and the number of outputs is 3

# Select loss function and optimizer.
model.compile(loss='categorical_crossentropy', optimizer='adam', metrics=['accuracy'])
```

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Configuration and Implementation of the Analysis Model (2)

- Training: 500 epochs

```
# Repeat training for the number
of epochs hist=model.fit(x_train, y_train, epochs=500, batch_size=20,validation_data=(x_valid, y_valid))
```

```
Epoch 1/500
6/6 [==========================================] - 0s 11ms/step - loss: 0.3215 - accuracy: 0.9583 - val_loss: 0.3842 - val_accuracy: 0.8333 Epoch 2/500 6/6

[==========================================] - 0s 4ms/step - loss: 0.3216 - accuracy: 0.9583 - val_loss: 0.3804 - val_accuracy: 0.8333 Epoch 3/500 6/6

[==============] - 0s 4ms/step - loss: 0.3207 - accuracy: 0.9583 - val_loss: 0.3792 - val_accuracy: 0.8333

    … …

Epoch 499/500
6/6 [============================] - 0s 4ms/step - loss: 0.1894 - accuracy: 0.9750 - val_loss: 0.2392 - val_accuracy: 0.9667 Epoch 500/500 6/6

[============================] - 0s 4ms/step - loss: 0.1892 - accuracy: 0.9750 - val_loss: 0.2395 - val_accuracy: 0.9333
```

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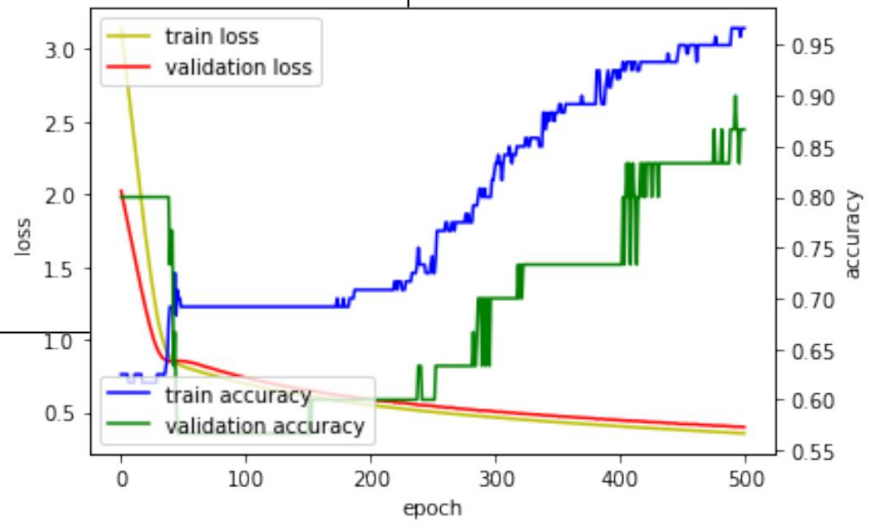
Configuration and Implementation of the Analysis Model (3)

- Visualization of training process and results

```
# Visualization of the training process

import matplotlib.pyplot as plt fig,
loss_ax = plt.subplots() acc_ax
= loss_ax.twinx()

loss_ax.plot(hist.history['loss'], 'y', label = 'train loss')
loss_ax.plot(hist.history['val_loss'], 'r', label = 'validation loss')

acc_ax.plot(hist.history['accuracy'], 'b', label = 'train accuracy')
acc_ax.plot(hist.history['val_accuracy'], 'g', label = 'validation accuracy')

loss_ax.set_xlabel('epoch')
loss_ax.set_ylabel('loss')
acc_ax.set_ylabel('accuracy')

loss_ax.legend(loc='upper left')
acc_ax.legend(loc='lower left')

plt.show()
```



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Performance evaluation of classification models

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Use of the Misclassification Table (1)

- In classification problems, there are often multiple categories/classes, but they can be divided into two as follows

  Ryuha has many binary classification problems.

  – Customer purchase status in target marketing

  – Good or bad credit status

  – Success or failure, etc.

- Creation of a misclassification table in binary classification

		Predicted		
		TRUE	FALSE	계
Actual	TRUE	O.K. (True Positive)	Missing (False Negative)	AT
	FALSE	False Alarm (False Positive)	O.K. (True Negative)	AF
	계	PT	PF	GS

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Use of the Misclassification Table (2)

- Creation of misclassification tables in binary classification and

Performance metrics

– In binary classification, to the model's prediction results

Confusion Matrix

composition

오분류표 (Confusion Matrix)				
		예측된 Class		합계
		Class = 1	Class = 0	
실제 Class	Class = 1	TP	FN	AT
	Class = 0	FP	TN	AF
합계		PT	PF	GS

- In addition to precision, sensitivity and specificity
  - Sensitivity: Recall of the TRUE class (Recall)
  - Specificity: Recall of the FALSE class (Recall)
- Accuracy using sensitivity and specificity

Calculation possible

– 정밀도 : $Class-specific\ Precision$

$P_P = TP/(TP+FP) = TP/PT$

$P_N = TP/(FN+TN) = TN/PF$

– 민감도 : $Sensitivity\ (True\ Positive\ Rate)$

$TPR = TP/(TP+FN) = TP/AT$

– 특이도 : $Specificity\ (True\ Negative\ Rate)$

$TNR = TN/(FP+TN) = TN/AF$

– 오류율 $= (FP+FN)/GS$, 정확도 $= (TP+TN)/GS$

– 정확도의 다른 계산 :

$Accuracy = Sensitivity \cdot (AT/GS) + Specificity \cdot (AF/GS)$

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Classification performance indicators (1)

- In classification, since the target variable is discrete, the target value and as follows

It can express performance metrics such as error rate and accuracy.

– Goal value (goal variable, label) : $y_i \in C, \ \ C = \{c_1, c_2, \cdots, c_p\}$

– Error rate = $\frac{1}{N}\sum_{i=1}^{N} I(\hat{y}_i \neq y_i)$

- **I(ÿ) is** a function that returns 1 if the condition inside the parentheses is true, otherwise returns 0.

– Accuracy = $\frac{1}{N}\sum_{i=1}^{N} I(\hat{y}_i = y_i) = 1 - \text{오류율}$

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Classification performance indicators (2)

• Calculation of accuracy and error rate as predicted results in multinomial classification –

Representation of classification results in the table below (general cases and examples)

		예측 (Predicted)			
		$\hat{c}_1$	$\hat{c}_2$	$\hat{c}_3$	계
실제(Actual)	c_1	n_{11}	n_{12}	n_{13}	n_1
	c_2	n_{21}	n_{22}	n_{23}	n_2
	c_3	n_{31}	n_{32}	n_{33}	n_3
	계	m_1	m_2	m_3	N

		예측 (Predicted)			
		A	B	C	계
실제(Actual)	A	21	5	1	27
	B	7	53	9	69
	C	3	12	16	31
	계	31	70	26	127

• In the example on the

right above – Number of correctly classified records: 90 (Accuracy:

90/127=0.7087) – Number of misclassified records: 37 (Error rate:

37/127=0.2913) • From the misclassification table on the left above, Precision and Recall used in the fields of pattern recognition or information retrieval

- 클래스별 정밀도 : $\mathrm{Prec}_i = n_{ii}/m_i$,
- 클래스별 재현율 : $Recall_i = n_{ii}/n_i$
- F-척도 : $F_i = 2/\left(\frac{1}{\mathrm{Prec}_i} + \frac{1}{Recall_i}\right) = \frac{2\mathrm{Prec}_i Recall_i}{\mathrm{Prec}_i + Recall_i} = \frac{2n_{ii}}{n_i + m_i}$